# ngspice-46 — Full Change Report

Every modification and its reason (original → current)

Ngspice + OpenVAF Enhancements project

July 2026

# Contents

## ngspice-46 — Full Change Report

Every modification applied to ngspice-46 in this project, with its reason. The baseline is the pristine ngspice-46 source tree (as vendored in the project's `original/` snapshot, which already contains OpenVAF-Reloaded's stock OSDI support); the current state is the tree committed in this repository under `ngspice-46/`. Each entry links the enhancement write-up in `enhancements_doc/` that carries the full engineering detail and the verifying example suite. The companion document openvaf_changes_full-report.md covers the compiler side.

> Maintenance note: this report is updated whenever an enhancement touches ngspice sources. If you are reading it alongside a newer tree, the per-enhancement index at the end tells you the last change it covers.

Scope summary. One new source file and 41 modified ones carry all the functional changes (~2,000 substantive diff lines). A further ~100 `Makefile.in` files and `config.h.in` differ only because the autotools were regenerated with a current libtool (E-77); they contain no hand-written changes and are not listed individually. Everything below is behavior, interface, or diagnostics.

---

### 1. The OSDI ABI, version 0.4 → 0.7

`src/osdi/osdi.h` is the authoritative C contract between ngspice and the `.osdi` objects `openvaf-r` produces. The project extended it four times — twice additively (new exported symbols old loaders simply never look up), twice with genuine version bumps (layout changes):

| Change | Kind | Enhancement | Why |
|---|---|---|---|
| `OSDI_ABSDELAY_COUNT` / `OSDI_ABSDELAY_INFOS` exports | additive symbols | E-1 | `absdelay()` needs simulator-side waveform history and synthetic delay-equation rows; the descriptors tell ngspice which nodes/offsets each delay slot uses |
| `OSDI_LAST_CROSSING_COUNT` / `OSDI_LAST_CROSSING_INFOS` exports | additive symbols | E-6 | `last_crossing()` needs waveform history and crossing interpolation — same additive pattern as absdelay |
| `OsdiNode.nodeset` field | ABI 0.5 (node stride change) | E-45 | Verilog-A net initializers (`electrical a = 5.0;`) become solver initial-guess nodesets; every node record carries the hint |
| `num_ac_stim_src` / `ac_stim_sources` / `load_ac_stim` descriptor tail | ABI 0.6 (descriptor grows) | E-51 | full `ac_stim()` support: a Verilog-A module can *be* the AC stimulus, so the descriptor must enumerate its AC right-hand-side sources |
| `load_noise` destination stride 1 → 2 (signed `(flat, react)` power pairs) | ABI 0.7 | E-54 | correct noise physics: operating-point-dependent factors and `ddt()`-shaped (jω) noise need a complex amplitude per source, `re + jω·im`, so coherent same-named sources (LRM 4.6.4) sum with correct sign and frequency shape |

Two additive flag conventions ride on the existing `eval()` interface (not version bumps):

- `EVAL_FLAG_IS_FINAL_STEP` (bit 1≪21) in the eval *input* flags — `@(final_step)` firing (E-53);
- eval *return* flags for `$finish`/`$stop` requests and `$discontinuity`-driven step rejection (E-55).

Pairing consequence: this ngspice requires OSDI $\geq$ 0.7 and rejects older objects with a clear version message; `.osdi` files from older compilers must be recompiled (E-54).

---

## 2. The OSDI runtime — `src/osdi/`

**`osdiaccept.c`** — new file (E-55, E-1, E-6)

The accepted-timepoint hook (`DEVaccept`) OSDI previously lacked. It advances the `absdelay/last_crossing` waveform histories only at *accepted* points (so rejected timesteps don't corrupt the delay lines) and reads the per-attempt latched `$finish/$stop` request flags at the one boundary where honoring them is safe. Why: acting on `$stop` mid-Newton-iteration was treated as a step failure and ground the timestep into a rejection loop; `$finish` was ignored outright.

**`osdidefs.h`** (104 diff lines)

- `OsdiExtraInstData` grows the per-instance fields behind every runtime feature: `dt/temp` offsets with given-flags (instance temperature), `eval_flags`, absdelay waveform-history rows and pre-allocated KLU/SPARSE matrix-pointer sets for the delay-equation rows (re-pointed on every DC$\leftrightarrow$AC transition, mirroring the regular Jacobian handling), and the last_crossing history (E-1, E-6, E-55).
- `ALIGN` macro renamed `OSDI_ALIGN`: the macOS SDK's `<sys/param.h>` defines an unrelated `ALIGN`, producing a redefinition warning in every OSDI translation unit (E-77).

**`osdiitf.h`** / **`osdiext.h`**

The registry-entry structure gains the absdelay/last_crossing descriptor pointers, and `OSDIpendingRequests()` is exported so the analyses can ask "did any device request `$finish/$stop` at this accepted point?" (E-1, E-6, E-55).

**`osdiregistry.c`** (68 diff lines)

- Loads the absdelay/last_crossing descriptor arrays from each `.osdi` and threads them into the registry entries (E-1, E-6).
- Requires OSDI $\geq$ 0.7 (E-54).

**`osdiinit.c`** (8 diff lines)

- Registers the `DEVaccept` hook (E-55).
- Publishes the synthetic instance-temperature parameter under both spellings, `dt` and the conventional `dtemp` every built-in device uses; `n1 a b mod dtemp=10` used to fail with *"unknown parameter"* while the underlying plumbing was exact (E-80).

**`osdiload.c`** (441 diff lines — the largest OSDI change)

- Stamps the absdelay synthetic rows (`y_synth`, `z`) for DC, AC and transient, driving the delay from the recorded history (E-1).
- Evaluates `last_crossing` from the recorded waveform with linear interpolation between the bracketing timepoints (E-6).
- Passes the final-step flag through to `eval()` (E-53).
- Latches eval-return flags per timepoint attempt (`point_eval_flags`, OR-ed across Newton iterations, reset per attempt) so `$finish/$stop` under solution-dependent conditions survive to the accepted-point boundary (E-55).

- Folds the stride-2 signed noise-power pairs (`fac·|fac|` semantics) when transferring `load_noise` results (E-54, E-42).

**`osdisetup.c`** (178 diff lines)

- Creates the synthetic delay-equation unknowns and matrix entries for absdelay (E-1).
- Applies the OSDI nodesets (`OsdiNode.nodeset`) as solver initial guesses (E-45).
- A `$fatal`/`$finish` raised during setup (models rejecting their configuration by design — HiSIM's `$port_connected` guards) used to surface as ngspice's baffling *"impossible error — can't occur"*; it now reads *"a Verilog-A device rejected its configuration during setup"* (E-56).

**`osdiacld.c`** (89 diff lines)

- AC loading gains the `ac_stim` right-hand-side injection: the partitioned AC-stimulus source array is loaded through `load_ac_stim` and added to the AC RHS with exact magnitude and phase, `$mfactor` applied linearly (deterministic signals scale with multiplicity) (E-51, E-26).
- Complex (jω-shaped) noise coupling entries participate in the AC matrix (E-54).

**`osdinoise.c`** (93 diff lines)

- Per-instance grouping of same-named noise sources: coherent summation of complex amplitudes `(a + jω·b)·T` before squaring, per LRM 4.6.4 — same-named sources correlate (including exact anti-phase cancellation), differently-named ones stay independent (E-42).
- Reads the stride-2 signed pairs of ABI 0.7 (E-54).

**`osditrunc.c`** (34 diff lines)

- Honors `$discontinuity(n)`'s next-step clamp: a negative `bound_step` sentinel from the model limits the step after the event (E-24).
- Honors the step-rejection return flag: when an event fired inside the step, `OSDItrunc` requests `delta/8` (with a `20·CKTdelmin` termination floor) so the integrator bisects onto the discontinuity instead of extrapolating across it (E-55).

**`osdi/Makefile.am`**

Adds `osdiaccept.c` to the build (E-55).

---

## 3. Analyses — `src/spicelib/analysis/`

**`dctran.c`** (46 diff lines)

- Advances OSDI delay/crossing histories via the accept path and skips history corruption on rejected steps (E-1, E-6).
- Issues the dedicated `@(final_step)` evaluation at the converged end of a successful transient (E-53).
- Honors accepted-point `$finish` (ends the analysis cleanly, firing `final_step` at the requesting point), `$stop` (pauses resumably), and aborts with a clear error on `$fatal`'s `E_PANIC` instead of retrying it as nonconvergence (E-55).

**`dcop.c`** (9) and **`acan.c`** (10)

- Fire `@(final_step)` at their successful end (an operating point fires both step events — a single point is first *and* last) (E-53).

- An AC job's operating-point phase carries the analysis name bit (only the name — adding the reactive `CALC_*` bits would wrongly enable integration at an op), so `analysis("ac")` holds through the whole AC run per LRM 4.6.1 and `@(initial_step("ac"))` fires in AC (E-53).

**`dctrcurv.c`** (150) + **`include/ngspice/trcvdefs.h`** (13)

- Generic **`.dc @inst[param]`** sweeps (a new sweep code): the DC sweep previously hardcoded Vsource/Isource/Resistor/temperature, so sweeping any other device parameter was a fatal error. The generic sweep resolves `@inst[param]` through the device's own parameter tables, refreshes per point via `DEVtemperature` (for OSDI exactly the `alter` path), nests with other sweep variables, and restores the original value afterwards (E-62).
- Fires `final_step` at the last sweep point; honors a `$finish` raised mid-sweep (E-53, E-55).

**`noisean.c`** (37)

- A singular AC matrix during noise analysis crashed ngspice with a SIGABRT: the code ignored `NIacIter`'s return and the adjoint solve asserted on the unfactored matrix. It now aborts cleanly ("AC solution failed at ... Hz") (E-56).
- Honors deferred `$finish`/`$stop` raised at the noise operating point and fires `final_step` on success (E-55, E-53).

**`span.c`** (31) + **`cktspdum.c`** (14)

- 1-port S-parameter analyses enabled: the "we need at least two ports" error was over-strict (a 1-port is a plain reflection measurement; the matrix machinery is N-general) — and it was hiding the `cadjoint` crash fixed in `dense.c` (E-64).
- **`Rbase`** published into the sp plot from port 1's `z0`, making the Touchstone writers work with no manual `let Rbase = 50` incantation (E-64).
- Stock NaN fixed in **`donoise`** noise-parameter extraction: the uncorrelated noise conductance `Gu` is analytically zero for fully-correlated single-source topologies, and floating-point rounding could land `sqrt(Ycor² + Gu/Rn)`'s argument at $-10^{-18} \rightarrow$ NaN for the noise figure. Clamped to the physical range $\geq 0$ — found by parity testing against an OSDI resistor that produced the exact textbook NF where the built-in returned NaN (E-63).

**`cktdisto.c`** (16)

- `.disto` silently reported zero distortion for OSDI devices — the distortion kernel needs higher-order Taylor coefficients the OSDI ABI cannot provide (first derivatives only), and such devices were skipped without a word. A prominent warning now names each affected OSDI device type (E-62).

---

## 4. Frontend — `src/frontend/`

**`plotting/pyplot.c`** (new) + **`com_pyplot.c`** (new) + **`plotit.c`**, **`commands.c`** (E-94)

A new interactive command, **`pyplot`**, plots simulated vectors with matplotlib — a Python counterpart to `gnuplot`. `ft_pyplot()` (`plotting/pyplot.c`) mirrors `ft_gnuplot()`: it writes a `<file>.data` table and a `<file>.py` matplotlib script and `system()`s `python3` (synchronously for a PNG, backgrounded for an interactive window). A new `"pyplot"` device arm in `plotit()` routes to it, so it accepts the same plot expressions as `plot`/`gnuplot`; the `com_pyplot()` wrapper mirrors `com_gnuplot()`, and `pyplot` is registered in both command tables. Two `set` variables tune it: `pyplot_terminal=png` renders headless (Agg) to a PNG, and `pyplot_python` picks the interpreter (default `python3`). Purely additive (E-94). The output file base name is optional (E-95): `com_pyplot()` treats the first word as a file name only when it is not a plot expression (no (, and not a vector by vec_get()), otherwise it defaults to pyplot — so pyplot v(out) plots directly (the command's minimum argument count was lowered from 2 to 1). ft_pyplot() also grew stacked subplots (set pyplot_subplots=N → plt.subplots(nrows, 1, sharex=True), N traces per panel; vs stays the x-axis, so it is not a panel separator) and matplotlib style sheets (set pyplot_style=<name>, dark → dark_background, guarded) (E-98). The headless pyplot_terminal was extended from PNG-only to the vector export formats svg and pdf (set pyplot_terminal=svg/pdf → fig.savefig(<file>.<fmt>), matplotlib picking the writer from the extension), and a figure size control was added (set pyplot_figsize="W,H" → plt.subplots(..., figsize=(W, H)); quote the value so ngspice keeps the comma) (E-99).

**postcoms.c** (414 diff lines) + **postcoms.h** + **commands.c** (16)

The Touchstone I/O subsystem (E-64, E-72):

- **wrsnp** (with wrs2p dispatching to the same handler): Touchstone v1 for any port count — the classic 2-port S11 S21 S12 S22 column order preserved byte-identically, N $\geq$ 3 in the spec's row-major layout with at most four complex pairs per line;
- writer options wrsnp <file> [ri|ma|db] [s|y|z] [hz|khz|mhz|ghz], any combination in any order: MA/DB formats, Y/Z parameters (normalized to Rbase, Y·R / Z/R, per the v1 spec), frequency units — the option line reflects every choice;
- **rdsnp**: reads any Touchstone v1 file into a new plot with a Hz frequency scale and complex vectors matching the .sp plot's conventions (MA/DB converted back, Y/Z de-normalized, the 2-port column order handled, Rbase published so imports round-trip) — measured data diffs against simulation in one let expression.

Why: E-63 showed the S-parameter *analysis* was exact but its results could not leave ngspice in the industry-standard format (wrs2p demanded a never-created Rbase vector), and nothing could bring VNA data in.

**outitf.c** (16)

- Integer operating-point variables recorded per point: getSpecial masked with IF_REAL only, so an integer opvar saved with .save @n1[flag] produced garbage in transient vectors (E-32).
- The plot-memory guard's message printed %Id — a Windows-only printf length modifier that is undefined behavior elsewhere and rendered the message as the numberless "memory required (Id Bytes)". Now %zu: real byte counts on every platform (E-77).

**resource.c** (27)

ft_ckspace() — the *"approaching max data size"* warning (RSS > 95% of RSS + free RAM) — now honors **set no_mem_check** (the same opt-out the fatal output-size estimate in outitf.c respects; one variable governs both memory guards) and warns once per excursion above the threshold, re-arming below 90%, instead of repeating on every check through a long analysis (E-81).

**display.c** (1)

#undef NODEV before the local macro definition — the macOS SDK's <sys/param.h> defines an unrelated NODEV (E-77).

---

## 5. Parser and device registry — `src/spicelib/`

**`parser/inpgtok.c`** (10)

A **`.model`** card override silently dropped its first parameter when the type name and the opening parenthesis had no space between them (`modname devtype(p1=v1 …)`): `INPgetNetTok`'s scan did not treat `(` as a token boundary, so the type token swallowed the paren and the first parameter name. Found while verifying event-function counters whose model-card overrides mysteriously kept defaults (E-8).

**`parser/inpgmod.c`** (5)

`find_model_parameter()` dereferenced `*(device->numModelParms)`, which is NULL for built-ins that take no model cards (VCVS, CCCS, …) — any *referenced* `.model m vcvs()` card segfaulted, with no OSDI involved at all (an ordinary MOS instance sufficed). This was the true root of the long-documented "module named like a built-in crashes" gotcha. One NULL guard; both shapes now produce clean, located errors (E-76).

**`devices/dev.c`** (51)

- Duplicate device-type registration warned and skipped: `osdi_add_device()` appended every descriptor unconditionally, and the model-card lookup scans front-to-back — so a module name duplicated across two loaded libraries silently resolved to the *first* registration (loading an updated model library gave the stale device with no hint), and a module named like a built-in corrupted model creation. Registration now checks the table case-insensitively and skips duplicates loudly (E-76).
- **`pre_osdi`** path dedup: loading an already-loaded file notes it and skips — with the remedy stated: *"restart ngspice to load a recompiled file"* (E-76, E-81).

**`osdi/osdiparam.c`** (E-93)

Setting a fixed (**`localparam`**) OSDI parameter from the netlist used to be swallowed silently: openvaf exports every `localparam` as a parameter, its "given" flag is forced false, and the written value is overwritten by the parameter-init default — so `.model … N=8` on a frozen structural width parameter changed nothing, with no feedback. `OSDIparam`/`OSDImParam` now test the new `PARA_FLAG_FIXED` descriptor flag (set by openvaf, a free bit of the parameter `flags` field) and emit *"parameter 'N' is a fixed (localparam) value and cannot be set from the netlist; ignored"* instead of silently dropping it (E-93).

---

## 6. Maths — `src/maths/`

**`dense/dense.c`** (11 substantive lines; the file also carries a whitespace/line-ending normalization from editing)

**`cadjoint()`** had no 1×1 base case: its cofactor loop allocated 0×0 and then negative-sized minors, killing ngspice with `malloc: can't allocate -8 bytes` — the crash that had been hiding behind the "at least two ports" guard in `span.c`. The adjugate of `[a]` is `[1]` (the empty minor's determinant is 1 by convention), making `cinverse` of a 1×1 equal `1/a`; a 100 Ω load on a 50 Ω port now writes a proper `.s1p` with S11 = 1/3 exactly (E-64).

**`sparse/spfactor.c`** (6)

The Markowitz-product overflow guards compare a `double` against `LARGEST_LONG_INTEGER` (= `LONG_MAX`), which is not exactly representable as a double; the conversion is now explicit — identical semantics, intentional and warning-free (E-77).

### `KLU/klu_multiply.c` (16)

`SMPmultiply`'s KLU path passed `NULL` internal↔external ordering maps to `KLU_matrix_vector_multiply`, which dereferenced them unconditionally (`&NULL[n]` = `0x14` for `n = 5`) and segfaulted — so `.option linesearch` crashed under `.option klu`. The line search is the only caller of `SMPmultiply` under KLU, so this path had never been exercised. A NULL map now means the identity ordering (`ext = i + 1`, since the KLU CSR is 0-based and the RHS/Solution vectors are 1-based), making the KLU matrix-vector product — and hence the E-111 line-search residual merit — correct under KLU, with a merit sequence numerically identical to Sparse 1.3 (E-112).

---

## 7. Build system

### `xspice/icm/makedefs.in` (4)

The XSPICE codemodel (`.cm`) link rule hardcoded the pre-macOS-10.3 idiom `-bundle -flat_namespace -undefined suppress`, deprecated loudly by the modern linker on all 14 codemodel links (CI macOS builds included). Now `-bundle -undefined dynamic_lookup` — the modern spelling for plugins whose host symbols resolve at load time (E-77).

### Autotools regeneration (the ~100 `Makefile.in` files + `config.h.in`)

Regenerated by `./autogen.sh` with libtool 2.5.4 during the zero-warning work — a build tree whose `configure` predates libtool 2.4.7 selects the deprecated darwin `-undefined suppress` flag whenever `MACOSX_DEPLOYMENT_TARGET` is unset. No hand-written content (E-77).

---

## 8. Per-enhancement index

Every enhancement that touched ngspice, oldest first:

| Enhancement | ngspice files | One line |
|---|---|---|
| E-1 | osdi.h, osdidefs.h, osdiitf.h, osdiregistry.c, osdiload.c, osdisetup.c, dctran.c (+accept path) | `absdelay()`: synthetic delay rows + waveform history |
| E-6 | osdi.h, osdidefs.h, osdiitf.h, osdiregistry.c, osdiload.c | `last_crossing()`: history + interpolated crossings |
| E-8 | parser/inpgtok.c | `.model` card first-parameter drop fix |
| E-24 | osditrunc.c | `$discontinuity` next-step clamp |
| E-25 | osdi callbacks (get_simparams) | expose `analysis_name` + `simulator` simparams |
| E-32 | outitf.c | integer opvars recorded per point |
| E-42 | osdinoise.c | coherent same-named noise grouping (LRM 4.6.4) |
| E-45 | osdi.h (ABI 0.5), osdisetup.c | Verilog-A nodesets applied to the solver |
| E-51 | osdi.h (ABI 0.6), osdiacld.c | full `ac_stim` AC-RHS injection |
| E-53 | dctran.c, dcop.c, dctrcurv.c, acan.c, noisean.c, osdiload.c | `@(final_step)` + analysis-name phase bits |
| E-54 | osdi.h (ABI 0.7), osdinoise.c, osdiload.c, osdiacld.c, osdiregistry.c | node-free complex noise factors, stride-2 pairs |

| Enhancement | ngspice files | One line |
|---|---|---|
| E-55 | osdiaccept.c (new), osdiload.c, osditrunc.c, osdiinit.c, dctran.c, dctrcurv.c, osdiitf.h, Makefile.am | `$finish`/`$stop`/`$fatal` honored; `$discontinuity` step rejection |
| E-56 | noisean.c, osdisetup.c | noise SIGABRT fix; setup-rejection diagnostics |
| E-62 | dctrcurv.c, trcvdefs.h, cktdisto.c | generic `.dc @inst[param]` sweeps; `.disto` warning |
| E-63 | span.c | `donoise` NaN clamp (stock defect, found by parity) |
| E-64 | span.c, cktspdum.c, dense.c, postcoms.c/.h, commands.c | Touchstone export, auto-Rbase, 1-port `.sp`, `cadjoint` 1×1 |
| E-72 | postcoms.c/.h, commands.c | Touchstone round 2: MA/DB/Y/Z/units + `rdsnp` reader |
| E-76 | dev.c, inpgmod.c | duplicate-registration warning, load dedup, stock `.model` segfault fix |
| E-77 | osdidefs.h, display.c, outitf.c, spfactor.c, makedefs.in (+autotools regen) | zero-warning build, 33 → 0 |
| E-80 | osdiinit.c | `dtemp` instance-parameter alias |
| E-81 | resource.c, dev.c | memory-warning opt-out + once-per-excursion; reload-note hint |
| E-93 | osdi.h, osdiparam.c | warn (not silently ignore) when a netlist sets a `PARA_FLAG_FIXED` (`localparam`) OSDI parameter |
| E-94 | plotting/pyplot.c (new), com_pyplot.c (new), plotit.c, commands.c | new `pyplot` command — plot vectors with matplotlib (a Python `gnuplot`) |
| E-95 | com_pyplot.c, commands.c | make the `pyplot` output file name optional (defaults to `pyplot`) |
| E-98 | plotting/pyplot.c | `pyplot` stacked subplots (`pyplot_subplots=N`) + matplotlib style sheets (`pyplot_style`) |
| E-99 | plotting/pyplot.c | `pyplot` vector export formats (`pyplot_terminal=svg/pdf`) + figure size (`pyplot_figsize`) |
| E-110 | optdefs.h, tskdefs.h, cktsopt.c, cktntask.c | `.option errpreset=conservative\|moderate\|liberal` — one knob for a coordinated tolerance/robustness set; explicit options override regardless of order (`moderate` = historical defaults) |
| E-111 | optdefs.h, tskdefs.h, cktdefs.h, cktsopt.c, cktdojob.c, cktntask.c, cktdest.c, niiter.c | `.option linesearch` — globalized (damped) Newton via Armijo backtracking on a new KCL-residual merit `‖F‖ =‖G·x−b‖`; result-neutral, off by default |
| E-112 | maths/KLU/klu_multiply.c | KLU support for `.option linesearch` — `SMPmultiply`'s KLU path passed NULL ordering maps that `klu_matrix_vector_multiply` dereferenced (SIGSEGV); NULL now means identity ordering. Line search verified merit-identical KLU vs Sparse |

*(E-25's simparam exposure lives in the OSDI callback table populated at load time; its diff rides inside the `osdiload.c`/callbacks changes.)*

Every entry above is guarded by a verify script under `examples/` and was regression-checked against the full example arsenal, the compiler's integration suite, and the VA_TEST industry-model corpus at the time it landed.